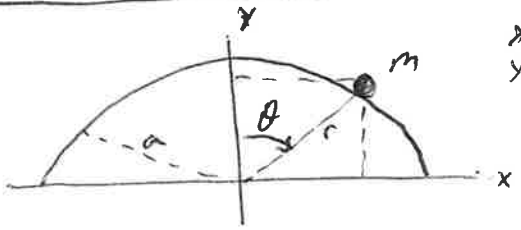
Problem Set #5 (SOLUTIONS)

1)



$x = r\sin\theta \qquad \dot{x} = \dot{r}\sin\theta + r\dot{\theta}\cos\theta$

$y = r\cos\theta \qquad \dot{y} = \dot{r}\cos\theta - r\dot{\theta}\sin\theta$

$f(r,\theta) = r - a = 0$

PART A

$L = T - V \qquad T = \frac{1}{2}m(\dot{x}^2+\dot{y}^2) = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2)$

$V = mgr\cos\theta$

$$\boxed{L(r,\dot{r},\theta,\dot{\theta}) = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - mgr\cos\theta}$$

PART B

$f(r,\theta) = r - a$ "equation of constraint", $\rightarrow \frac{\partial f}{\partial r} = 1 \qquad \frac{\partial f}{\partial \theta} = 0$

$$r:\ \frac{d}{dt}\frac{\partial L}{\partial \dot{r}} - \frac{\partial L}{\partial r} = \sum_j \lambda_j \frac{\partial f_j}{\partial r} \quad m\ddot{r} - (mr\dot{\theta}^2 - mg\cos\theta) = \lambda\frac{\partial f}{\partial r}$$

$$\theta:\ \frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = \sum_j \lambda_j \frac{\partial f_j}{\partial \theta} \quad mr^2\ddot{\theta} - mgr\sin\theta + 2mr\dot{r}\dot{\theta} = \lambda\cdot\frac{\partial f}{\partial \theta}$$

$$\Rightarrow \boxed{\begin{aligned} mr\dot{\theta}^2 - mg\cos\theta - m\ddot{r} + \lambda &= 0 \\ mgr\sin\theta - mr^2\ddot{\theta} - 2mr\dot{r}\dot{\theta} &= 0 \end{aligned}}$$

Apply the constraint: $r = a \rightarrow \dot{r} = \ddot{r} = 0$

$$\boxed{\begin{aligned} ma\dot{\theta}^2 - mg\cos\theta + \lambda &= 0 \\ mga\sin\theta - ma^2\ddot{\theta} &= 0 \end{aligned}} \rightarrow \boxed{\ddot{\theta} = \frac{g}{a}\sin\theta}$$

1) PART C (continued...)

$$\ddot{\theta} = \frac{d^2\theta}{dt^2} = \frac{d}{dt}\dot{\theta}[\theta(t)] = \frac{d\dot{\theta}}{d\theta}\left(\frac{d\theta}{dt}\right) = \dot{\theta}\left(\frac{d\dot{\theta}}{d\theta}\right)$$

$$\Rightarrow \dot{\theta}\frac{d\dot{\theta}}{d\theta} = \frac{g}{a}\sin\theta \rightarrow \int_0^{\dot{\theta}} \dot{\theta}\, d\dot{\theta} = \frac{g}{a}\int_0^{\theta} \sin\theta\, d\theta$$

$$\frac{\dot{\theta}^2}{2} = -\frac{g}{a}\cos\theta\Big|_0^{\theta} = \frac{g}{a}(\cos(0) - \cos\theta) = \frac{g}{a}(1-\cos\theta)$$

$$\boxed{\dot{\theta}^2 = \frac{2g}{a}(1-\cos\theta)}$$

PART D

$$mg\left[\frac{2g}{a}(1-\cos\theta)\right] - mg\cos\theta + \lambda = 0$$

$$\Rightarrow \lambda = mg\cos\theta - 2mg(1-\cos\theta)$$

$$\lambda = 3mg\cos\theta - 2mg \Rightarrow \boxed{\lambda = mg(3\cos\theta - 2)}$$

Constraint forces:

$$\boxed{Q_r^{(r)} = \sum_j \lambda_j \frac{\partial f_j}{\partial r} = \lambda\frac{\partial f}{\partial r} = mg(3\cos\theta - 2)}$$

$$\boxed{Q_\theta^{(r)} = \sum_j \lambda_j \frac{\partial f_j}{\partial \theta} = 0}$$

PART E

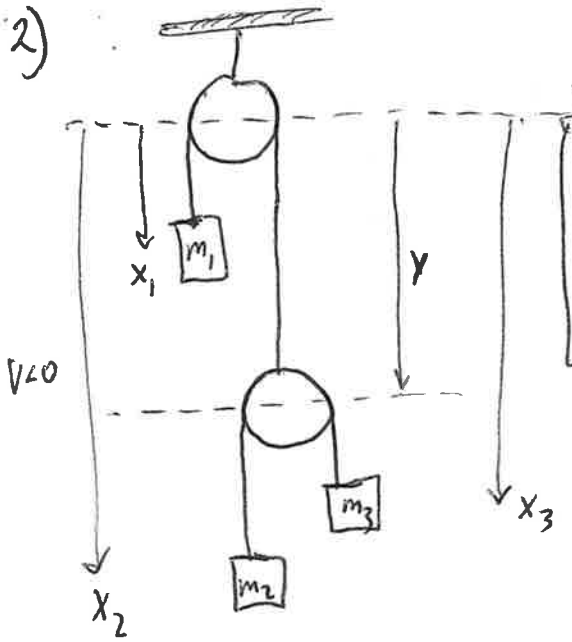
At $\theta \rightarrow \theta_0$, $Q_r^{(r)} \rightarrow 0 \Rightarrow mg(3\cos\theta_0 - 2) = 0 \Rightarrow \cos\theta_0 = \frac{2}{3}$ or $\boxed{\theta_0 = \cos^{-1}\left(\frac{2}{3}\right)}$

$$\boxed{\theta_0 = 48.1^\circ}$$

PART F

For $\theta \rightarrow 0$ (top), $Q_r^{(r)} = mg(3\cos(0) - 2) \Rightarrow \boxed{Q_r^{(r)} = mg}$ ($\cos(0) = 1$)

2)



PART A $\quad L = T - V$

$$L = \frac{1}{2}\left(m_1\dot{x}_1^2 + m_2\dot{x}_2^2 + m_3\dot{x}_3^2\right) + g\left(m_1x_1 + m_2x_2 + m_3x_3\right)$$

$$\Rightarrow L = L(x_j, \dot{x}_j),\ j = 1, 2, 3$$

PART B

$$l_1 = x_1 + y \qquad l_2 = (x_2 - y) + (x_3 - y) \Rightarrow l_2 = x_2 + x_3 - 2y$$

$$l_2 = x_2 + x_3 - 2(l_1 - x_1)$$

$$x_2 + x_3 - 2l_1 - l_2 + 2x_1 = 0$$

$$\Rightarrow \boxed{2x_1 + x_2 + x_3 - (2\cdot l_1 + l_2) = 0}$$

↳ Eq. of constraint $f(x_1, x_2, x_3) = 0$

$$\frac{d^2f}{dt^2} \equiv \ddot{f} \Rightarrow \boxed{2\ddot{x}_1 + \ddot{x}_2 + \ddot{x}_3 = 0}$$

PART C

$$x_1:\ \frac{d}{dt}\frac{\partial L}{\partial \dot{x}_1} - \frac{\partial L}{\partial x_1} = \lambda\frac{\partial f}{\partial x_1} \longrightarrow \boxed{m_1\ddot{x}_1 - m_1g = 2\lambda}$$

$$x_2:\ \frac{d}{dt}\frac{\partial L}{\partial \dot{x}_2} - \frac{\partial L}{\partial x_2} = \lambda\frac{\partial f}{\partial x_2} \longrightarrow \boxed{m_2\ddot{x}_2 - m_2g = \lambda}$$

$$x_3:\ \frac{d}{dt}\frac{\partial L}{\partial \dot{x}_3} - \frac{\partial L}{\partial x_3} = \lambda\frac{\partial f}{\partial x_3} \longrightarrow \boxed{m_3\ddot{x}_3 - m_3g = \lambda}$$

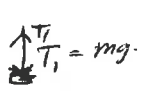
PART D

$$\left.\begin{aligned}\ddot{x}_1 &= (2\lambda + m_1g)/m_1\\ \ddot{x}_2 &= (\lambda + m_2g)/m_2\\ \ddot{x}_3 &= (\lambda + m_3g)/m_3\end{aligned}\right\} \quad 2\ddot{x}_1 + \ddot{x}_2 + \ddot{x}_3 = \frac{2(2\lambda + m_1g)}{m_1} + \frac{(\lambda + m_2g)}{m_2} + \frac{(\lambda + m_3g)}{m_3} = 0$$

$$\Rightarrow \frac{4\lambda}{m_1} + \frac{2g}{m_1} + \frac{\lambda}{m_2} + g + \frac{\lambda}{m_3} + g = 0$$

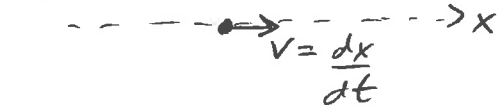
$$\lambda\left(\frac{4}{m_1} + \frac{1}{m_2} + \frac{1}{m_3}\right) + 4g = 0 \Rightarrow \boxed{\lambda = \frac{-4g}{\left(\frac{4}{m_1} + \frac{1}{m_2} + \frac{1}{m_3}\right)}}$$

2) PARTE



$$Q_1^{(r)} = \cancel{2\lambda_1} \; m_1 g - m_1 \ddot{x}_1 = -2\lambda = \boxed{\frac{8g}{\frac{4}{m_1} + \frac{1}{m_2} + \frac{1}{m_3}}}$$

3) $F(x,t) = \frac{k}{x^2} e^{-t/\tau}$, $k, \tau > 0$



1D motion

PART A

$L = T - V$ $\quad T = \frac{1}{2} m \dot{x}^2$, $\quad dV = -F(x,t)\,dx = -\int k e^{-t/\tau} x^{-2} dx$

$$V = \frac{-k e^{-t/\tau} x^{-2+1}}{-2+1} + V_0 \quad (V_0 \text{ arb. const.})$$

$$\Rightarrow V = \frac{k}{x} e^{-t/\tau}$$

$$\boxed{L = \frac{1}{2} m \dot{x}^2 - \frac{k}{x} e^{-t/\tau}}$$

PART B

$H = T + V = \sum_\sigma p_\sigma \dot{q}_\sigma - L$

$\underbrace{m\dot{x}}_{p_x}(\dot{x}) - L = p_x \dot{x} - \frac{1}{2}\frac{m^2\dot{x}^2}{m} + \frac{k}{x} e^{-t/\tau}$

$= m\dot{x}^2 - \frac{1}{2} m \dot{x}^2 + \frac{k}{x} e^{-t/\tau}$

$H = \underbrace{m\dot{x}^2}_{\frac{1}{2}\frac{m^2\dot{x}^2}{m} \Rightarrow \frac{p_x^2}{2m}} + \frac{k}{x} e^{-t/\tau}$

$$\boxed{H = \frac{p_x^2}{2m} + \frac{k}{x} e^{-t/\tau} = T + V}$$

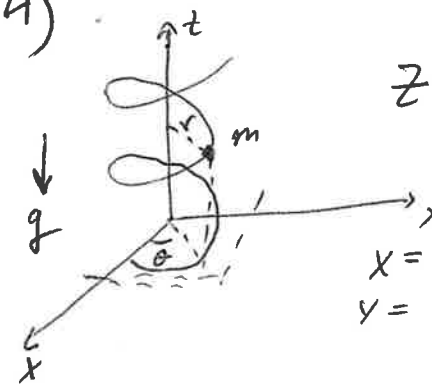
PART C

$H = H(p_x, q_x, t) \rightarrow$

$$\boxed{\frac{dH}{dt} = \frac{k}{x} e^{-t/\tau}\left(-\frac{1}{\tau}\right) \neq 0}$$

$H \rightarrow$ evolves with time, and hence NOT conserved.

4)



$z = k\theta$, $r = \text{const.}$, $k = \text{const.}$

$x = r\cos\theta$, $y = r\sin\theta$

$\dot{x} = -r\dot{\theta}\sin\theta$

$\dot{y} = r\dot{\theta}\cos\theta$

$\dot{z} = k\dot{\theta}$

PART A

$H = T + V$, $T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) = \frac{1}{2}m(r^2\dot{\theta}^2 + k^2\dot{\theta}^2)$

$V = mgz$

$$\boxed{H = \frac{1}{2}m(r^2+k^2)\dot{\theta}^2 + mgk\theta}$$

Alternatively, $H = \sum_\sigma p_\sigma \dot{q}_\sigma - L$,

$L = T - V = \frac{1}{2}m(r^2\dot{\theta}^2 + k^2\dot{\theta}^2) - mgz$ (in terms of z)

($\dot{\theta}^2 = \dot{z}^2/k^2$, $k^2\dot{\theta}^2 = \dot{z}^2$)

$L = \frac{1}{2}m\left(\frac{r^2}{k^2}+1\right)\dot{z}^2 - mgz$, $p_z = \frac{\partial L}{\partial \dot{z}} = m\left[\frac{r^2}{k^2}+1\right]\dot{z}$

$\Rightarrow H = p_z\dot{z} - \frac{1}{2}m\left[\frac{r^2}{k^2}+1\right]\dot{z}^2 + mgz$

($m\left[\frac{r^2}{k^2}+1\right]\dot{z} = p_z$)

$= \frac{1}{2}p_z\dot{z} + mgz$

$$\boxed{H(p_z, z) = \frac{1}{2}\frac{p_z^2}{m\left[\frac{r^2}{k^2}+1\right]} + mgz}$$

4) PART B

$$\dot{z} = \frac{\partial H}{\partial p_z} = \frac{p_z}{m\left[\frac{r^2}{k^2}+1\right]}$$

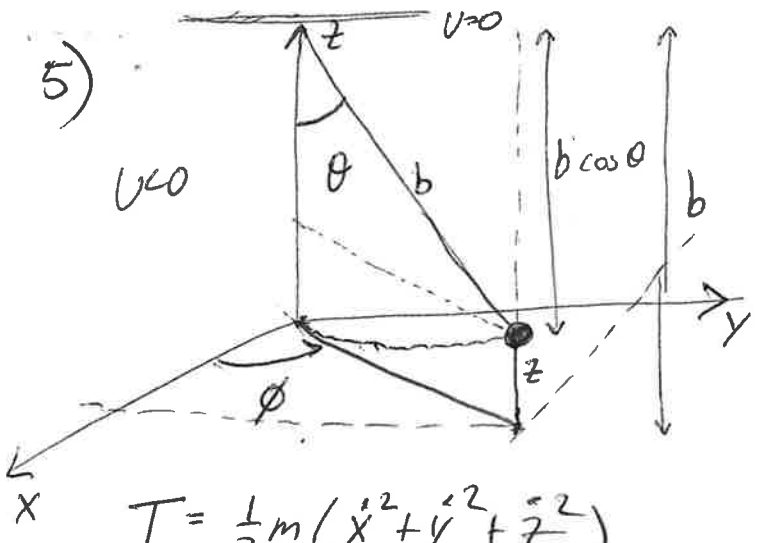
$$-\dot{p}_z = \frac{\partial H}{\partial z} = mg$$

$$\ddot{z} = \frac{\dot{p}_z}{m\left[\frac{r^2}{k^2}+1\right]} = -\frac{mg}{m\left[\frac{r^2}{k^2}+1\right]}$$

PART C

$$\ddot{z} = -\frac{g}{\left[\frac{r^2}{k^2}+1\right]}$$

5)



$$x = b\sin\theta\cos\phi$$
$$y = b\sin\theta\sin\phi$$
$$z = b(1-\cos\theta)$$

$$\dot{x} = b\dot{\theta}\cos\theta\cos\phi - b\dot{\phi}\sin\theta\sin\phi$$
$$\dot{y} = b\dot{\theta}\cos\theta\sin\phi + b\dot{\phi}\sin\theta\cos\phi$$
$$\dot{z} = \dot{\theta} b\sin\theta$$

$$T = \frac{1}{2}m(\dot{x}^2+\dot{y}^2+\dot{z}^2)$$
$$= \frac{1}{2}m(b^2\dot{\theta}^2\cos^2\theta + b^2\dot{\phi}^2\sin^2\theta + \dot{\theta}^2 b^2\sin^2\theta)$$
$$T = \frac{1}{2}m(b^2\dot{\theta}^2 + b^2\dot{\phi}^2\sin^2\theta), \quad V = -mgb\cos\theta$$

$$H(p_\theta, p_\phi, \theta, \phi) = \sum_{\sigma=\theta,\phi} p_\sigma \dot{q}_\sigma - L = p_\theta\dot{\theta} + p_\phi\dot{\phi} - L$$

$$L = T - V = \frac{1}{2}mb^2(\dot{\theta}^2 + \dot{\phi}^2\sin^2\theta) + mgb\cos\theta$$

$$p_\sigma = \frac{\partial L}{\partial \dot{q}_\sigma} \rightarrow p_\theta = \frac{\partial L}{\partial\dot{\theta}} = mb^2\dot{\theta}, \quad p_\phi = \frac{\partial L}{\partial\dot{\phi}} = mb^2\dot{\phi}\sin^2\theta$$

$$H = p_\theta\dot{\theta} + p_\phi\dot{\phi} - \underbrace{\frac{1}{2}mb^2\dot{\theta}^2}_{p_\theta\dot{\theta}} - \underbrace{\frac{1}{2}mb^2\dot{\phi}^2\sin^2\theta}_{p_\phi\dot{\phi}} - mgb\cos\theta$$

$$H = \frac{1}{2}p_\theta\dot{\theta} + \frac{1}{2}p_\phi\dot{\phi} - mgb\cos\theta$$

Now need to convert $(\dot{\theta}\ \dot{\phi}) \rightarrow p_\theta, p_\phi \rightarrow \dot{\theta} = \frac{p_\theta}{mb^2}$ $\dot{\phi} = \frac{p_\phi}{mb^2\sin^2\theta}$

$$\boxed{H = \frac{1}{2}\frac{p_\theta^2}{mb^2} + \frac{1}{2}\frac{p_\phi^2}{mb^2\sin^2\theta} - mgb\cos\theta} \Rightarrow H(p_\theta, p_\phi, \theta\phi)$$

5) PART B

$$\dot{\theta} = \frac{\partial H}{\partial p_\theta} = \frac{p_\theta}{mb^2} \qquad \dot{\phi} = \frac{\partial H}{\partial p_\phi} = \frac{p_\phi}{mb^2 \sin^2\theta}$$

$$\dot{p}_\theta = -\frac{\partial H}{\partial \theta} = \frac{p_\phi^2 \cos\theta}{mb^2 \sin^3\theta} - mgb\sin\theta$$

$$\dot{p}_\phi = -\frac{\partial H}{\partial \phi} = 0$$